

BEIRUT RAFIC HARIRI INTL, LebanonWMO: **401000**Lat: **33.821N**Lon: **35.488E**Elev: **87**StdP: **14.65**Time Zone: **2.00 (LBN)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	47.1	49.7	26.8	20.8	56.8	30.6	24.8	57.3	31.7	57.9	27.7	58.2	7.8	110	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	91.1	74.1	88.7	75.7	87.6	76.0	80.4	87.1	79.2	86.0	78.5	85.4	8.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.6	148.7	86.0	77.2	141.9	85.1	76.2	137.1	84.4	44.2	87.0	43.0	86.4	42.1	85.5	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.9	18.8	15.9	DB	43.5	98.0	3.0	3.1	41.3	100.2	39.6	102.1	37.9	103.8	35.7	106.1	(4)
(5)			WB	37.6	82.7	3.3	1.5	35.3	83.8	33.3	84.7	31.4	85.5	29.0	86.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.7	58.2	58.8	62.1	66.7	72.5	77.6	81.7	83.0	81.0	76.1	68.5	61.5	(6)	
	DBStd	9.63	3.92	4.32	5.01	5.28	4.28	2.69	1.86	1.70	2.25	3.98	4.51	4.00	(7)	
	HDD50	4	2	2	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	685	212	179	120	33	1	0	0	0	0	1	18	121	(9)	
	CDD50	7559	257	248	374	501	696	829	983	1024	930	808	554	355	(10)	
	CDD65	2764	2	5	29	84	232	379	518	559	480	343	122	11	(11)	
	CDH74	24650	6	16	160	450	1097	2910	5734	6673	4910	2313	365	16	(12)	
	CDH80	7269	0	2	47	150	287	558	1837	2476	1406	458	47	1	(13)	
(14) Wind	WSAvg	6.8	7.0	7.2	7.2	7.0	6.7	7.1	7.6	6.6	6.3	5.8	5.8	6.7	(14)	
(15) Precipitation	PrecAvg	30.9	7.1	5.0	4.2	1.8	0.6	0.0	0.0	0.0	0.2	2.0	4.3	6.3	(15)	
	PrecMax	45.6	25.2	12.7	10.4	9.1	3.1	0.4	0.2	0.4	1.2	6.8	12.4	13.0	(16)	
	PrecMin	13.9	2.1	1.3	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	(17)	
	PrecStd	8.5	3.9	2.8	2.3	1.7	0.7	0.1	0.0	0.1	0.3	1.6	2.8	3.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.5	76.6	85.9	91.2	93.5	91.2	91.8	92.7	91.4	91.7	84.6	75.7	(19)	
		MCWB	57.6	57.9	60.8	62.7	66.2	69.2	78.4	80.1	76.7	68.1	64.7	59.9	(20)	
	2%	DB	68.3	70.5	78.0	84.1	87.4	86.6	88.9	89.8	88.2	87.4	80.7	71.9	(21)	
		MCWB	57.4	57.6	59.2	62.4	66.5	73.8	78.4	79.4	75.8	69.6	64.7	59.6	(22)	
	5%	DB	66.3	67.7	73.2	78.8	82.4	84.4	87.5	88.1	86.5	84.2	77.6	69.8	(23)	
		MCWB	56.7	57.4	58.8	62.6	67.4	74.1	77.7	78.2	75.2	70.6	64.3	59.1	(24)	
	10%	DB	64.5	65.3	69.4	74.8	79.0	82.7	86.2	87.4	85.8	82.3	75.4	67.9	(25)	
		MCWB	55.5	56.4	59.4	62.7	68.5	73.7	77.1	77.7	74.9	70.6	63.8	58.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.9	63.0	65.4	68.7	74.1	78.9	81.7	82.4	80.7	77.2	70.9	64.5	(27)	
		MCDB	67.4	69.2	72.9	77.7	80.3	85.5	88.0	88.9	88.0	83.1	78.3	71.3	(28)	
	2%	WB	59.5	60.6	63.5	66.9	72.2	77.3	80.0	80.7	78.6	75.0	68.7	62.1	(29)	
		MCDB	65.9	66.8	70.3	75.2	78.3	83.1	86.4	87.6	85.8	82.0	76.7	69.3	(30)	
	5%	WB	57.9	59.0	61.9	65.5	71.0	76.2	79.1	79.6	77.4	73.4	66.9	60.3	(31)	
		MCDB	64.6	65.5	68.8	73.3	77.2	82.1	85.5	86.6	84.9	80.9	75.2	67.5	(32)	
	10%	WB	56.5	57.3	60.3	64.2	69.8	75.0	78.3	78.8	76.2	71.9	65.0	59.0	(33)	
		MCDB	63.2	64.0	67.6	71.8	76.2	81.2	84.8	85.9	84.1	80.2	73.5	66.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	11.6	12.5	13.3	12.8	11.3	10.5	10.1	10.2	11.0	11.8	11.1	(35)	
		MCDBR	13.0	15.1	19.2	20.8	18.2	13.5	12.1	11.3	11.4	13.3	14.1	13.4	(36)	
		MCWBR	10.0	11.2	11.1	11.0	10.8	10.4	10.2	9.5	10.0	11.4	11.9	10.4	(37)	
	5% WB	MCDBR	11.8	13.3	15.3	15.9	13.7	12.2	11.4	11.2	10.9	11.2	12.4	11.8	(38)	
		MCWBR	9.8	10.9	10.8	10.3	10.4	10.6	10.1	9.9	10.5	11.0	11.4	9.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.400	0.440	0.493	0.549	0.541	0.462	0.479	0.488	0.467	0.482	0.424	0.406	(40)	
	taud		2.209	2.080	1.942	1.825	1.885	2.134	2.098	2.070	2.117	2.052	2.180	2.200	(41)	
	Ebn at Noon		249	251	247	240	244	263	258	253	252	234	238	238	(42)	
	Edh at Noon		36	46	56	67	63	49	51	51	47	46	37	35	(43)	
(44) All-Sky Solar Radiation	RadAvg		806	1069	1533	1939	2336	2585	2528	2290	1930	1413	1011	763	(44)	
	RadStd		76	116	100	130	107	43	35	44	51	72	65	75	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.77	N/A	-2.61	+0.92	N/A	N/A	N/A	-124	+409	+281	(46)
(47)	Regional (13 neighbors)	+0.54	N/A	-2.39	N/A	N/A	N/A	N/A	N/A	+311	+198	(47)

Nomenclature: See separate page